UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Solutions of Homogeneous Linear DEs: Complementary Functions

Complete study pack for class 002: the auxiliary equation; real distinct, real repeated and complex roots; combinations for equations of order n; solving with initial and boundary conditions

Compiled from the PESU Unit 2 class 002 slide deck and AV-summary transcript, the Unit 1 & 2 class-work sheet (problems 1–9) and eight past ESA papers. Generated 15 October 2026.

· How to use this pack

Topic 1 derives the auxiliary equation, topic 2 gives the three second-order cases, topic 3 extends them to order n, topic 4 works every class-work problem, including those with initial conditions. Learn Figure F3.1 (one CF block per root) by heart: every later Unit 2 class (particular integrals, variation of parameters, Legendre and Cauchy equations, LCR circuits, springs) starts by writing this CF. Table C4.2 shows the CF of every Unit 2 ESA question. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Unit 2 deck 002, slide 7 (rendered .pptx)
T002 00:09:10	Unit 2 AV-summary transcript 002 (“Rules for finding the Complementary Function, Problems”), segment start
CW U2 P8	Unit 2 class-work problem 8
ESA Jan-24 s4 2c	Past paper: UE23MA141A Jan 2024 set 4, question 2.c

Contents

1. 1 · The auxiliary equation and the complementary function
2. 2 · Second-order equations: three cases
3. 3 · Equations of order n: combining the cases
4. 4 · Solving problems, with initial conditions
5. Appendix A · Symbol glossary
6. Appendix B · Sources, coverage and gaps

1 The auxiliary equation and the complementary function

A homogeneous linear DE with constant coefficients, $f(D)y = 0$, is solved by trying $y = e^{mx}$. Substituting gives $f(m)e^{mx} = 0$, and since $e^{mx} \neq 0$ the exponent m must satisfy the algebraic equation $f(m) = 0$, the *auxiliary equation* (AE, also called the characteristic equation). Its general solution is called the *complementary function* (CF). The form of the CF depends only on the nature of the roots of the AE.

Key terms and definitions

Homogeneous linear DE with constant coefficients. $(D^n + k_1 D^{n-1} + k_2 D^{n-2} + \cdots + k_n)y = 0$, written $f(D)y = 0$, where $D = \frac{d}{dx}$ and $k_1, \ldots, k_n$ are constants. S002·10 · T002 00:12:58

Auxiliary polynomial. $f(m)$: replace D by m in the linear differential operator $f(D)$. S002·6 · T002 00:04:00

Auxiliary equation (characteristic equation). $f(m) = 0$. For $\frac{d^2y}{dx^2} + k_1\frac{dy}{dx} + k_2 y = 0$ it is $m^2 + k_1 m + k_2 = 0$. S002·5 · S002·6 · T002 00:03:01 · T002 00:04:23

Complementary function (CF). The general solution of the homogeneous linear DE $f(D)y = 0$. It contains as many arbitrary constants as the order of the equation. T002 00:00:41 · T002 00:06:31

> *"Substituting into the above equation, we see that $y = e^{mx}$ is a solution if $(m^2 + k_1 m + k_2)e^{mx} = 0$. $m^2 + k_1 m + k_2 = 0$ is the AUXILIARY EQUATION of $(D^2 + k_1 D + k_2)y = 0$."* S002·5

> *"The solution of the differential equation depends on the nature of roots of the auxiliary equation."* S002·6

Formulas

E1.1 · Trial solution and its derivatives ESA ★★

$$y = e^{mx}, \qquad y' = m\,e^{mx}, \qquad y'' = m^2 e^{mx}$$

gives	Turns $f(D)y = 0$ into $f(m)e^{mx} = 0$.
applies	Any linear DE with constant coefficients.
note	$D^k e^{mx} = m^k e^{mx}$, so $f(D)e^{mx} = f(m)e^{mx}$.

S002·4 · T002 00:02:04

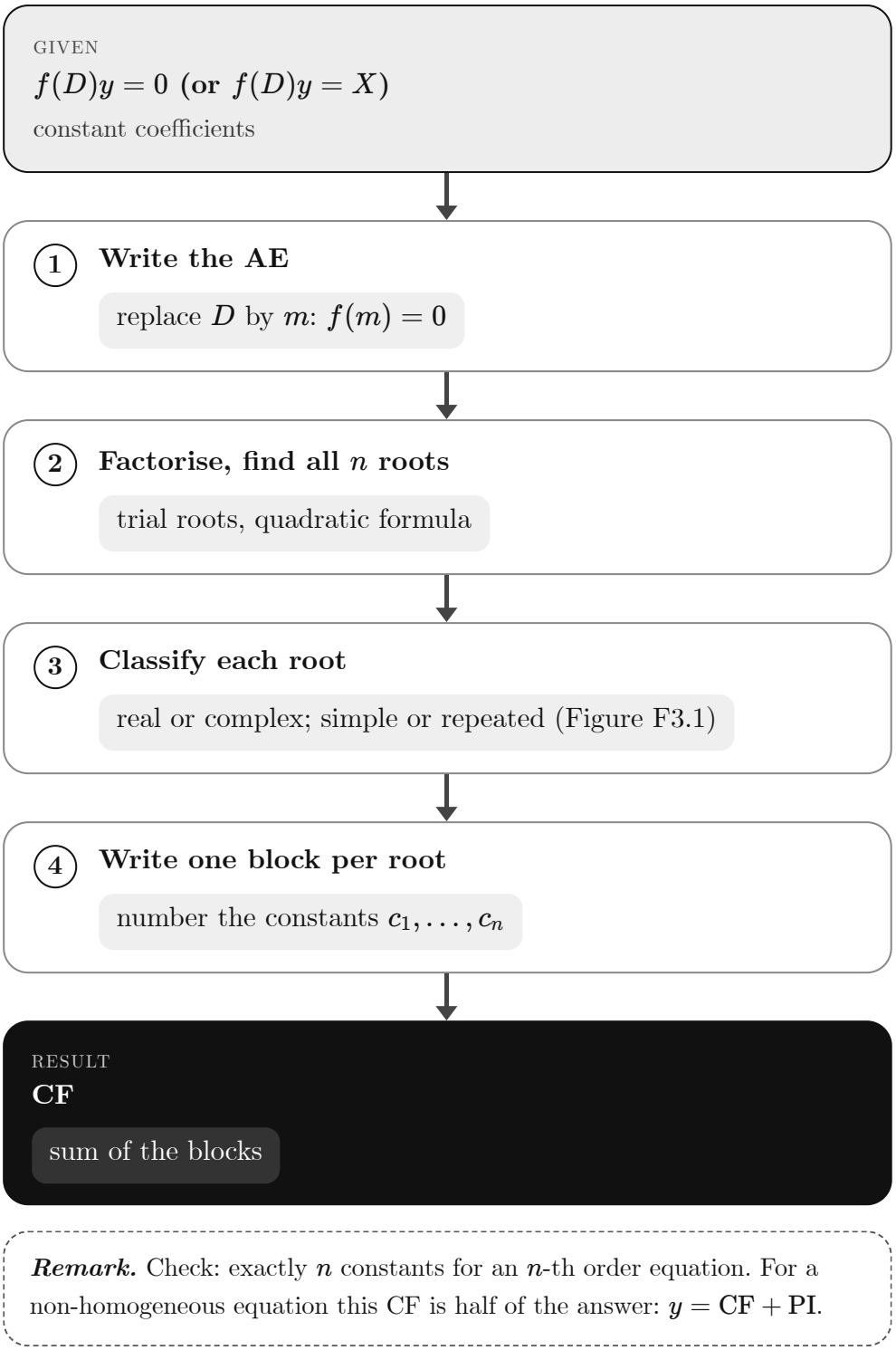
E1.2 · Auxiliary equation ESA ★★★

$$f(D)y = 0 \implies f(m) = m^n + k_1 m^{n-1} + \cdots + k_n = 0$$

gives	The n roots $m_1, \ldots, m_n$ that fix the CF.
applies	Homogeneous, constant coefficients.
note	Replace D by m (equivalently $y \to 1$, $y' \to m$, $y'' \to m^2$, $y''' \to m^3$).

S002·5 · S002·10 · T002 00:03:18 · T002 00:07:40

Flowcharts



Remark. Check: exactly n constants for an n-th order equation. For a non-homogeneous equation this CF is half of the answer: $y = \mathrm{CF} + \mathrm{PI}$.

Figure F1.1. Finding the complementary function ESA ★★★

T002 00:03:18 · T002 00:07:40 · T002 00:08:07 · T002 00:20:41

Important points and why

1. Every non-homogeneous ESA question in Unit 2 (24 of 24 occurrences) needs the CF first; a slip in the AE loses the whole question. ESA ★★★ ESA Dec-23 3a · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c

2. Writing the AE: replace D by m in $f(D)$, or in the DE replace y by 1, $\frac{dy}{dx}$ by m, $\frac{d^2y}{dx^2}$ by m^2, $\frac{d^3y}{dx^3}$ by m^3, and equate to 0. ESA ★★★ T002 00:03:36 · T002 00:07:40
3. $e^{mx} \neq 0$ for every real x, so the bracket $f(m)$ must be 0. That is why only the roots of the AE matter. ESA ★★ S002·5 · T002 00:02:47
4. An equation of order n has n arbitrary constants in its general solution; count them as a check. ESA ★★★ T002 00:06:31 · T002 00:15:54

Why does the CF matter when the exam only asks non-homogeneous equations?
The general solution of $f(D)y = X$ is $y = \mathbf{CF} + \mathbf{PI}$. The CF is the general solution of $f(D)y = 0$, so every question in classes 003–015 starts here.

T002 00:00:41 · ESA Dec-23 3a

Exam trap. The AE for $\frac{d^3y}{dx^3}$ needs m^3. The lecture says "replace it by m square" at T002 00:07:49; that is a slip of the tongue. ADDED FOR CLARITY T002 00:07:49

Exam trap. A missing y term means a zero root: $m^3 + 4m = m(m^2 + 4)$. Do not divide by m and lose the constant c_1. ADDED FOR CLARITY S002·14

Exam trap. With a leading coefficient, keep it in the factorisation: $4m^2 - 8m + 3 = (2m - 1)(2m - 3)$ gives $m = \frac{1}{2}, \frac{3}{2}$ (CW P4). ADDED FOR CLARITY CW U2 P4

SHORT-NOTE RECAP

For $f(D)y = 0$ with constant coefficients, put $y = e^{mx}$: $f(D)e^{mx} = f(m)e^{mx} = 0$, and $e^{mx} \neq 0$, so $f(m) = 0$, the auxiliary (characteristic) equation. Write it by replacing D by m (or $y \to 1$, $y' \to m$, $y'' \to m^2$). Its roots decide the form of the complementary function, the general solution of the homogeneous equation, which has as many arbitrary constants as the order.

2 Second-order equations: three cases

For $y'' + k_1 y' + k_2 y = 0$ the AE $m^2 + k_1 m + k_2 = 0$ has two roots. Case I, real and distinct: $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$. Case II, real and equal: $y = (c_1 + c_2 x) e^{mx}$. Case III, complex $\alpha \pm i\beta$: $y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$. Every second-order CF has exactly two arbitrary constants.

Key terms and definitions

> *"CASE I. If the roots m_1 and m_2 are real and distinct, then the general solution of the differential equation $\frac{d^2y}{dx^2} + k_1 \frac{dy}{dx} + k_2 y = 0$ is $y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$."* S002·7

> *"CASE II. If the roots m_1 and m_2 are real and equal, then the general solution of the differential equation $\frac{d^2y}{dx^2} + k_1 \frac{dy}{dx} + k_2 y = 0$ is $y = (c_1 + c_2 x) e^{mx}$."* S002·8

> *"CASE III. If the roots m_1 and m_2 are complex, let the roots be $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$; then the general solution of the differential equation is $y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$."* S002·9

Formulas

E2.1 · Case I: real and distinct roots ESA ★★★

$$m_1 \neq m_2 \text{ real}: \quad y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

gives	The CF.
applies	Discriminant $k_1^2 - 4k_2 > 0$.
note	Slide 002·7: $m = -3, 1$ gives $y = c_1 e^{-3x} + c_2 e^{x}$.

S002·7 · T002 00:05:40

E2.2 · Case II: real and equal roots ESA ★★★

$$m_1 = m_2 = m: \quad y = (c_1 + c_2 x) e^{mx}$$

gives	The CF.
applies	$k_1^2 - 4k_2 = 0$, $m = -\frac{k_1}{2}$.
note	"This is the formula you should remember" (T002 00:09:10).

S002·8 · T002 00:08:57 · T002 00:09:10

E2.3 · Case III: complex roots

ESA ★★★

$$m = \alpha \pm i\beta : \quad y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$$

GIVES The CF, in real form.

APPLIES $k_1^2 - 4k_2 < 0$; α = real part, β = imaginary part ($\beta > 0$).

NOTE $\alpha = 0$ (roots $\pm i\beta$) gives $c_1 \cos \beta x + c_2 \sin \beta x$.

S002·9 · T002 00:10:56

E2.4 · Roots of the quadratic AE

ESA ★★

$$m = \frac{-k_1 \pm \sqrt{k_1^2 - 4k_2}}{2}$$

GIVES Which case applies, from the sign of the discriminant.

APPLIES AE $m^2 + k_1 m + k_2 = 0$.

NOTE Lecture: "$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$" with $a = 1, b = -4, c = 13$ for slide 002·9.

ADDED FOR CLARITY T002 00:11:56

Graphs and diagrams

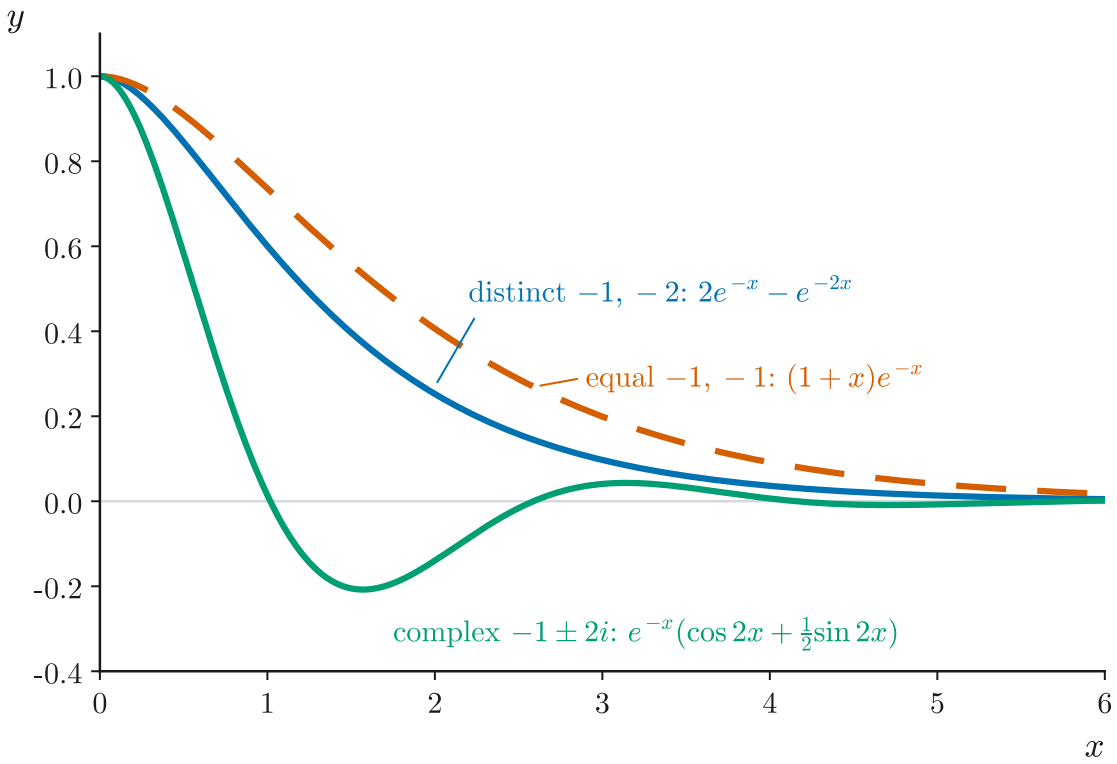


Figure G2.1. The three second-order cases from the same start. ESA ★★★

ADDED FOR CLARITY S002·7 · S002·8 · S002·9

How to read it. Each curve solves a second-order equation with $y(0) = 1$, $y'(0) = 0$. Distinct roots $-1, -2$ ($y'' + 3y' + 2y = 0$, blue) and the double root -1 ($y'' + 2y' + y = 0$, vermilion dashed) decay without crossing zero; the double root is the fastest return that does not oscillate. Complex roots $-1 \pm 2i$ ($y'' + 2y' + 5y = 0$, green) oscillate inside the decaying envelope e^{-x}. Each solution was found and checked with sympy.

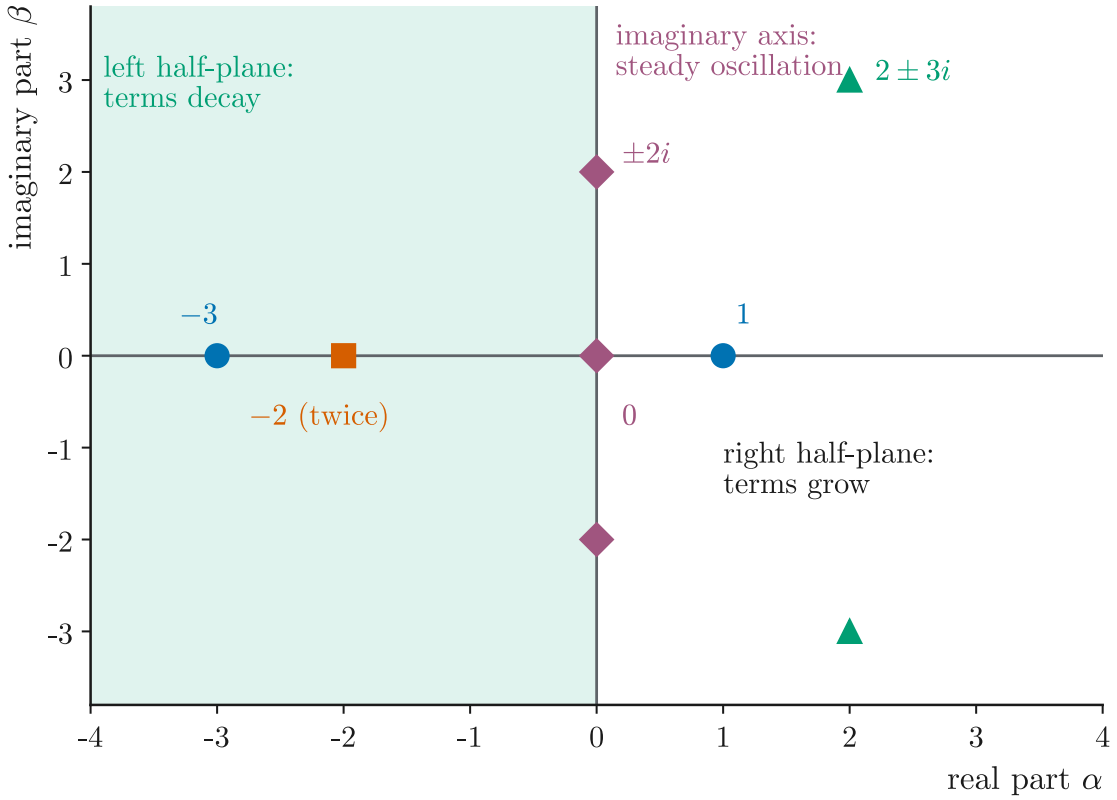
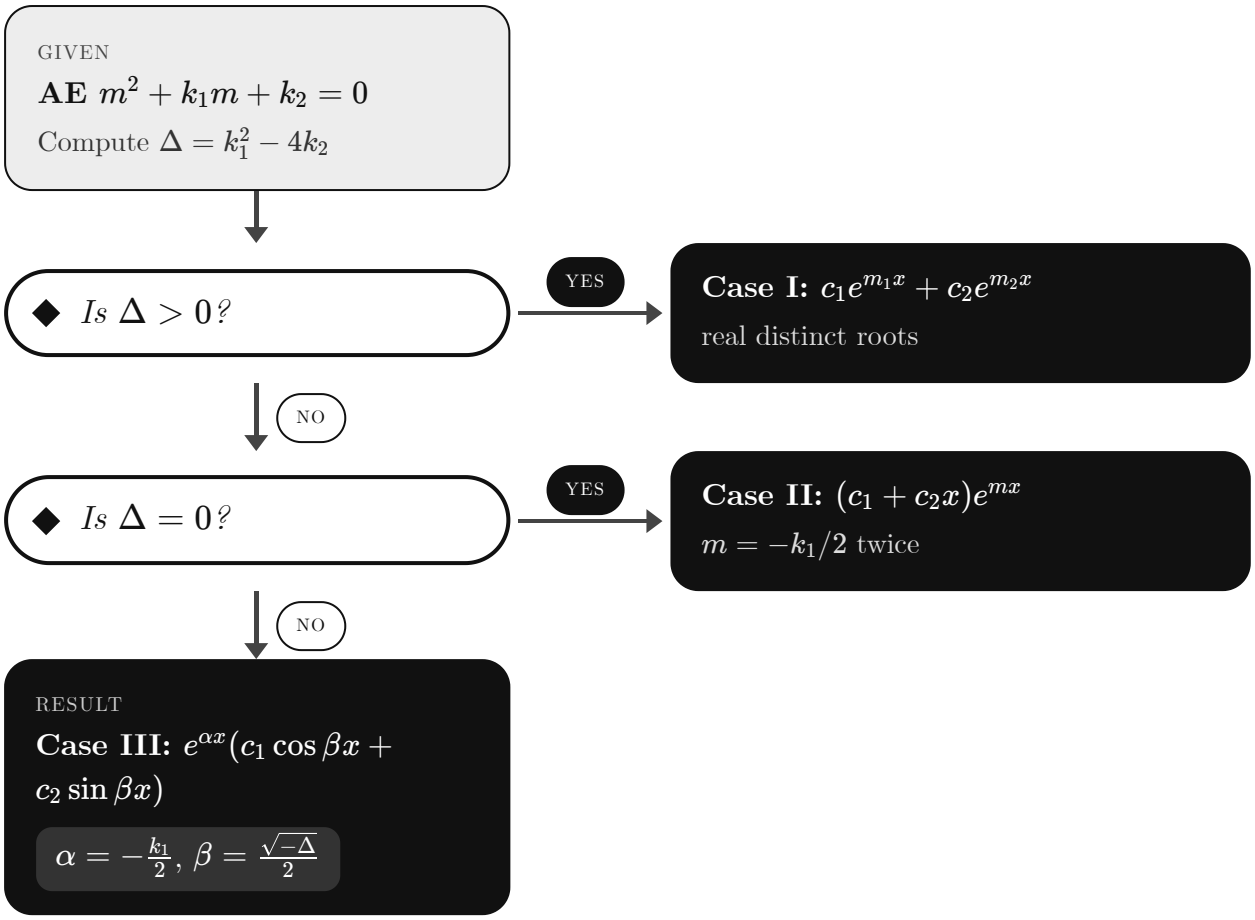


Figure G2.2. Where the roots lie decides the shape of the solution. ESA ★★

ADDED FOR CLARITY S002·7 · S002·8 · S002·9 · S002·14

How to read it. The roots of the four slide examples in the complex plane: $-3, 1$ (slide 002·7, circles), -2 twice (slide 002·8, square), $2 \pm 3i$ (slide 002·9, triangles), $0, \pm 2i$ (slide 002·14, diamonds). The real part α gives growth ($\alpha > 0$) or decay ($\alpha < 0$, green half-plane); the imaginary part β gives oscillation $\cos \beta x$, $\sin \beta x$. Roots on the real axis give no oscillation.

Flowcharts



Remark. The discriminant test is added for clarity; the slides classify the roots after solving the AE.

Figure F2.1. Which case? A second-order equation ESA ★★★

ADDED FOR CLARITY S002·7 · S002·8 · S002·9 · T002 00:11:56

The three second-order cases

Roots of AE	CF	Slide example	Answer	Shape (Figure G2.1)
real, distinct $m_1 \neq m_2$	$c_1 e^{m_1 x} + c_2 e^{m_2 x}$	$y'' + 2y' - 3y = 0$	$c_1 e^{-3x} + c_2 e^{x}$	sum of exponentials
real, equal m	$(c_1 + c_2 x)e^{mx}$	$y'' + 4y' + 4y = 0$	$(c_1 + c_2 x)e^{-2x}$	no oscillation
complex $\alpha \pm i\beta$	$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$	$y'' - 4y' + 13y = 0$	$e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$	oscillates; envelope $e^{\alpha x}$

S002·7 · S002·8 · S002·9

Important points and why

1. Classify the roots before writing anything: real and distinct, real and equal, or complex. The quadratic formula settles it. ESA ★★★ S002·6 · T002 00:04:41 · T002 00:11:56
2. In case III, α is the real part and β the imaginary part of the root: $2 \pm 3i$ gives $e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$. ESA ★★★ S002·9 · T002 00:12:13
3. Complex roots of an AE with real coefficients occur in conjugate pairs $\alpha \pm i\beta$, so they always come two at a time. ESA ★★ T002 00:10:38

Why $(c_1 + c_2x)e^{mx}$ and not $c_1e^{mx} + c_2e^{mx}$ for equal roots?
$c_1e^{mx} + c_2e^{mx} = (c_1 + c_2)e^{mx}$ has only one independent constant, but a second-order equation needs two. xe^{mx} is a second, independent solution: for $(D - m)^2y = 0$, $(D - m)(xe^{mx}) = e^{mx}$ and $(D - m)e^{mx} = 0$.
ADDED FOR CLARITY S002·8 · T002 00:06:31

Where do $\cos \beta x$ and $\sin \beta x$ come from?
$e^{(\alpha \pm i\beta)x} = e^{\alpha x}(\cos \beta x \pm i \sin \beta x)$ (Euler's formula). Suitable combinations of the two complex solutions give the real solutions $e^{\alpha x} \cos \beta x$ and $e^{\alpha x} \sin \beta x$.
ADDED FOR CLARITY S002·9

Exam trap. Do not write $c_1e^{mx} + c_2e^{mx}$ for a double root: it is $(c_1 + c_2x)e^{mx}$. ADDED FOR CLARITY S002·8

Exam trap. For $2 \pm 3i$, $\beta = 3$, not 2. The lecture first says "beta is 2", then corrects it to 3 (T002 00:12:13); slide 002·9 has $\cos 3x$, $\sin 3x$. S002·9 · T002 00:12:13

Exam trap. Do not put i in the CF: $e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$, never $e^{(2+3i)x}$ terms.
ADDED FOR CLARITY S002·9

Exam trap. Roots $\pm 2i$ have $\alpha = 0$: the CF is $c_1 \cos 2x + c_2 \sin 2x$, with no e^{2x}.
ADDED FOR CLARITY S002·14

Worked examples

W2.1 · Case I: $y'' + 2y' - 3y = 0$ (slide 002·7) ESA ★★★

GIVEN Solve $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - 3y = 0$.

STEP 1 $(D^2 + 2D - 3)y = 0$; AE $m^2 + 2m - 3 = 0$.

STEP 2 $(m + 3)(m - 1) = 0$, so $m = -3, 1$: real and distinct.

STEP 3 E2.1.

ANSWER $\boxed{y = c_1 e^{-3x} + c_2 e^{x}}$

CHECK sympy dsolve and checkodesol agree.

S002·7 · T002 00:07:04 · T002 00:08:11

W2.2 · Case II: $y'' + 4y' + 4y = 0$ (slide 002·8, CW P2) ESA ★★★

GIVEN Solve $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$.

STEP 1 AE $m^2 + 4m + 4 = 0$, i.e. $(m + 2)^2 = 0$.

STEP 2 $m = -2, -2$: real and equal.

STEP 3 E2.2 with $m = -2$.

ANSWER $\boxed{y = (c_1 + c_2 x)e^{-2x}}$

CHECK sympy agrees. CW P2 is the same equation, with the same answer.

S002·8 · T002 00:09:27 · T002 00:10:03 · CW U2 P2

W2.3 · Case III: $y'' - 4y' + 13y = 0$ (slide 002·9) ESA ★★★

GIVEN Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 13y = 0$.

STEP 1 AE $m^2 - 4m + 13 = 0$.

STEP 2 $m = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$, so $\alpha = 2$, $\beta = 3$.

STEP 3 E2.3.

ANSWER $\boxed{y = e^{2x}(c_1 \cos 3x + c_2 \sin 3x)}$

CHECK sympy agrees.

S002·9 · T002 00:11:38 · T002 00:12:27

W2.4 · $y'' - 4y' - 12y = 0$ (CW P1) ESA ★★

GIVEN Solve $y'' - 4y' - 12y = 0$.

STEP 1 AE $m^2 - 4m - 12 = 0$, i.e. $(m-6)(m+2) = 0$.

STEP 2 $m = 6, -2$: real and distinct.

ANSWER $\boxed{y = c_1 e^{6x} + c_2 e^{-2x}}$

CHECK CW answer agrees; sympy agrees.

CW U2 P1

W2.5 · CF of an LCR circuit (ESA Jan-24 s3 3b, CF part) ESA ★★★

GIVEN $R = 40\,\Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F: the charge satisfies $q'' + 40q' + 625q = 100\cos 5t$. Find the CF.

STEP 1 $\frac{1}{LC} = \frac{1}{16\times 10^{-4}} = 625$, so AE $m^2 + 40m + 625 = 0$.

STEP 2 $m = \frac{-40 \pm \sqrt{1600 - 2500}}{2} = -20 \pm 15i$.

ANSWER $\boxed{q_{\text{CF}} = e^{-20t}(c_1 \cos 15t + c_2 \sin 15t)}$

CHECK sympy roots agree. The PI and the full answer are in the class 014 (LCR) pack; the CF is the decaying transient.

ESA Jan-24 s3 3b

Past-paper questions

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q3.b (CF part) · solved in W2.5

Series circuit: $R = 40\,\Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F, $E = 100\cos 5t$. Find the charge at any time t. (CF: complex roots.)

ESA Dec 2023 · 7 marks · UE22MA141A Q3.a (CF part; also May 2024 Q3.b, June 2024 Q2.a) · solved in W2.2

Solve $(D-2)^2 y = 8(e^{2x} + \sin 2x)$. (CF: $(c_1 + c_2 x)e^{2x}$, a double root.)

SHORT-NOTE RECAP

$y'' + k_1 y' + k_2 y = 0$, AE $m^2 + k_1 m + k_2 = 0$. Real distinct m_1, m_2: $c_1 e^{m_1 x} + c_2 e^{m_2 x}$ ($-3, 1$: $c_1 e^{-3x} + c_2 e^{x}$). Real equal m: $(c_1 + c_2 x)e^{mx}$ ($-2, -2$: $(c_1 + c_2 x)e^{-2x}$). Complex $\alpha \pm i\beta$: $e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$ ($2 \pm 3i$: $e^{2x}(c_1 \cos 3x + c_2 \sin 3x)$).

3 Equations of order n: combining the cases

An n-th order equation has an AE of degree n, so n roots and n arbitrary constants. Each root, or pair of complex roots, contributes its own block to the CF. Each block is multiplied by $1, x, x^2, \ldots$, one power for each time the root is repeated. Slides 002·11–13 list five cases (all distinct, all equal, four equal, two complex pairs, a repeated complex pair). Every CF you will write in Unit 2 is a combination of these cases.

Key terms and definitions

Multiplicity of a root. The number of times a root is repeated in the AE: in $(m-1)^2(m+1)^2=0$ both 1 and -1 have multiplicity 2. (added for clarity) S002·12 · T002 00:14:47

> *"Case (i). If the roots $m_1, m_2, \ldots, m_n$ are real and distinct, then the general solution is $y = c_1e^{m_1x} + c_2e^{m_2x} + \cdots + c_ne^{m_nx}$."* S002·11

> *"Case (ii). If the roots $m_1, m_2, \ldots, m_n$ are real and equal (say equal to m), then the general solution is $y = (c_1 + c_2x + \cdots + c_nx^{n-1})e^{mx}$."* S002·12

> *"Case (iii). If the roots m_1, m_2, m_3, m_4 are real and equal (say equal to m) and the remaining $(n-4)$ roots are real and distinct, then the general solution is $y = (c_1 + c_2x + c_3x^2 + c_4x^3)e^{mx} + c_5e^{m_5x} + c_6e^{m_6x} + \cdots + c_ne^{m_nx}$."* S002·12

> *"Case (iv). If there are two pairs of imaginary roots $\alpha \pm i\beta$, $\gamma \pm i\delta$, then the general solution is $y = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x) + e^{\gamma x}(c_3\cos\delta x + c_4\sin\delta x) + c_5e^{m_5x} + \cdots + c_ne^{m_nx}$."* S002·13

> *"Case (v). If a pair of complex roots say $\alpha \pm i\beta$ occurring twice, then the GS is $y = e^{\alpha x}\big((c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x\big) + c_5e^{m_5x} + \cdots + c_ne^{m_nx}$."* S002·13

Formulas

E3.1 · Case (i): n real distinct roots ESA ★★★

$$y = c_1e^{m_1x} + c_2e^{m_2x} + \cdots + c_ne^{m_nx}$$

gives	The CF of an n-th order equation.
applies	All roots real and different.
note	n constants.

S002·11 · T002 00:14:12

E3.2 · Case (ii): a root repeated k times

ESA ★★★

$$y = (c_1 + c_2x + c_3x^2 + \cdots + c_kx^{k-1})e^{mx}$$

GIVES The block of the CF for a k-fold real root m.

APPLIES Slide 002·12 has $k = n$ (all roots equal) and $k = 4$.

NOTE Stop at x^{k-1}: a double root gives $c_1 + c_2x$, a triple root $c_1 + c_2x + c_3x^2$.

S002·12 · T002 00:14:58 · T002 00:15:15

E3.3 · Case (iii): equal roots with distinct ones

ESA ★★

$$y = (c_1 + c_2x + c_3x^2 + c_4x^3)e^{mx} + c_5e^{m_5x} + \cdots + c_ne^{m_nx}$$

GIVES The CF when four roots are equal and the rest distinct.

APPLIES $m_1 = m_2 = m_3 = m_4 = m$.

NOTE Add the blocks of each root.

S002·12 · T002 00:16:12 · T002 00:16:37

E3.4 · Case (iv): two pairs of complex roots

ESA ★★

$$y = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x) + e^{\gamma x}(c_3\cos\delta x + c_4\sin\delta x) + c_5e^{m_5x} + \ldots$$

GIVES The CF.

APPLIES Roots $\alpha \pm i\beta$, $\gamma \pm i\delta$, rest real and distinct.

NOTE Each pair has its own two constants (CW P5).

S002·13 · T002 00:17:08 · T002 00:17:36

E3.5 · Case (v): a complex pair occurring twice

ESA ★★

$$y = e^{\alpha x}\left[(c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x\right] + c_5e^{m_5x} + \ldots$$

GIVES The CF.

APPLIES $\alpha \pm i\beta$ repeated twice.

NOTE Both the cosine and the sine get a polynomial of degree 1 (CW P7).

S002·13 · T002 00:18:21 · T002 00:18:51

E3.6 · A complex pair occurring three times ESA ★

$$y = e^{\alpha x}\left[(c_1 + c_2 x + c_3 x^2)\cos\beta x + (c_4 + c_5 x + c_6 x^2)\sin\beta x\right] + \dots$$

GIVES The CF.

APPLIES $\alpha \pm i\beta$ repeated three times.

NOTE Said in the lecture; not on the slides.

TRANSCRIPT ONLY T002 00:19:14 · T002 00:19:32 · T002 00:19:45

E3.7 · Zero root ESA ★★

$$m = 0: \quad c\,e^{0x} = c$$

GIVES A constant term in the CF.

APPLIES $f(m)$ has a factor m (no y term in the DE).

NOTE Slide 002·14: $m = 0, \pm 2i$ gives $c_1 + c_2\cos 2x + c_3 \sin 2x$. A double zero root gives $c_1 + c_2 x$.

S002·14 · T002 00:21:20

Graphs and diagrams

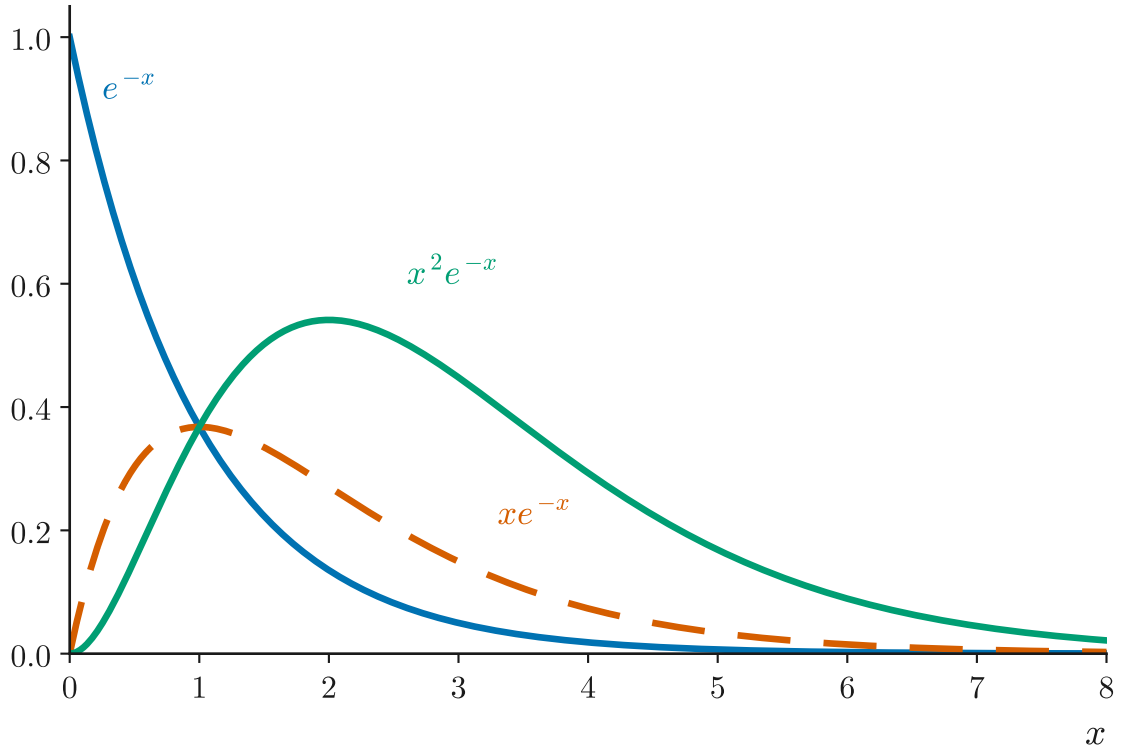


Figure G3.1. A triple root $m = -1$: the three solutions $x^k e^{-x}$. ESA ★★

ADDED FOR CLARITY S002·12 · T002 00:15:15

How to read it. For $(D+1)^3 y = 0$ the CF is $(c_1 + c_2 x + c_3 x^2)e^{-x}$. The three curves e^{-x}, xe^{-x}, x^2e^{-x} have different shapes (peaks at $x = 0, 1, 2$), so they are independent and give three constants. All three still decay because $m < 0$.

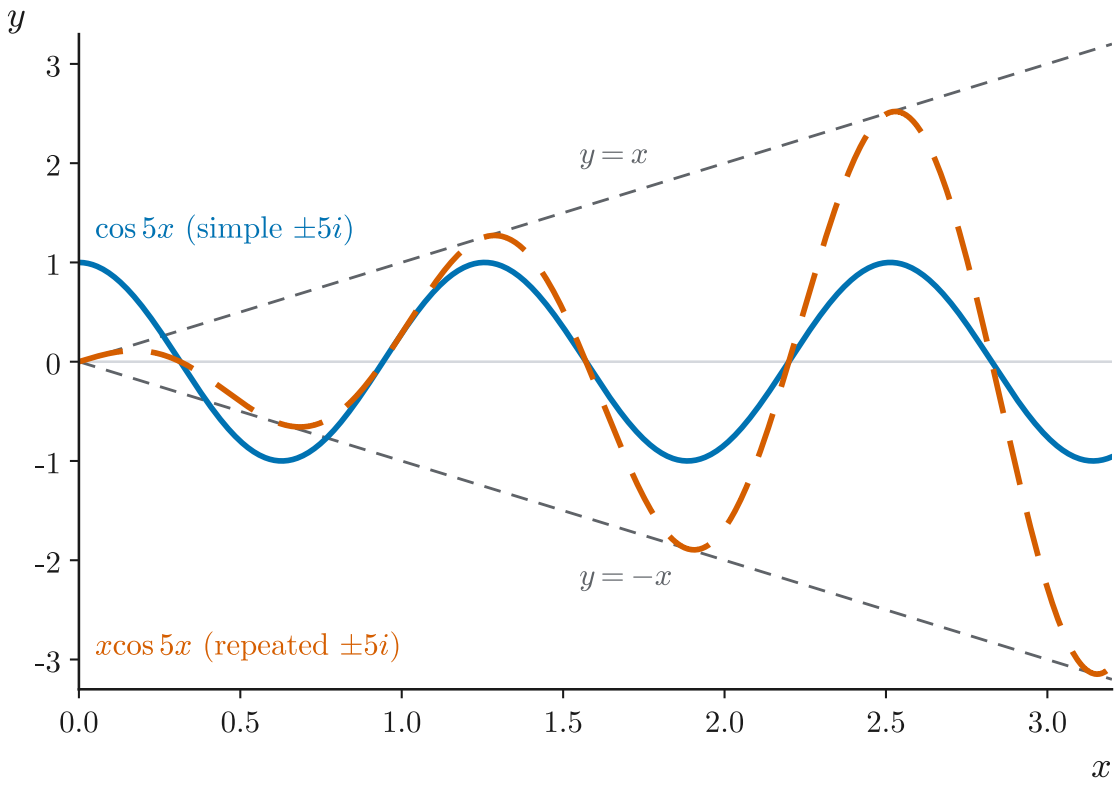
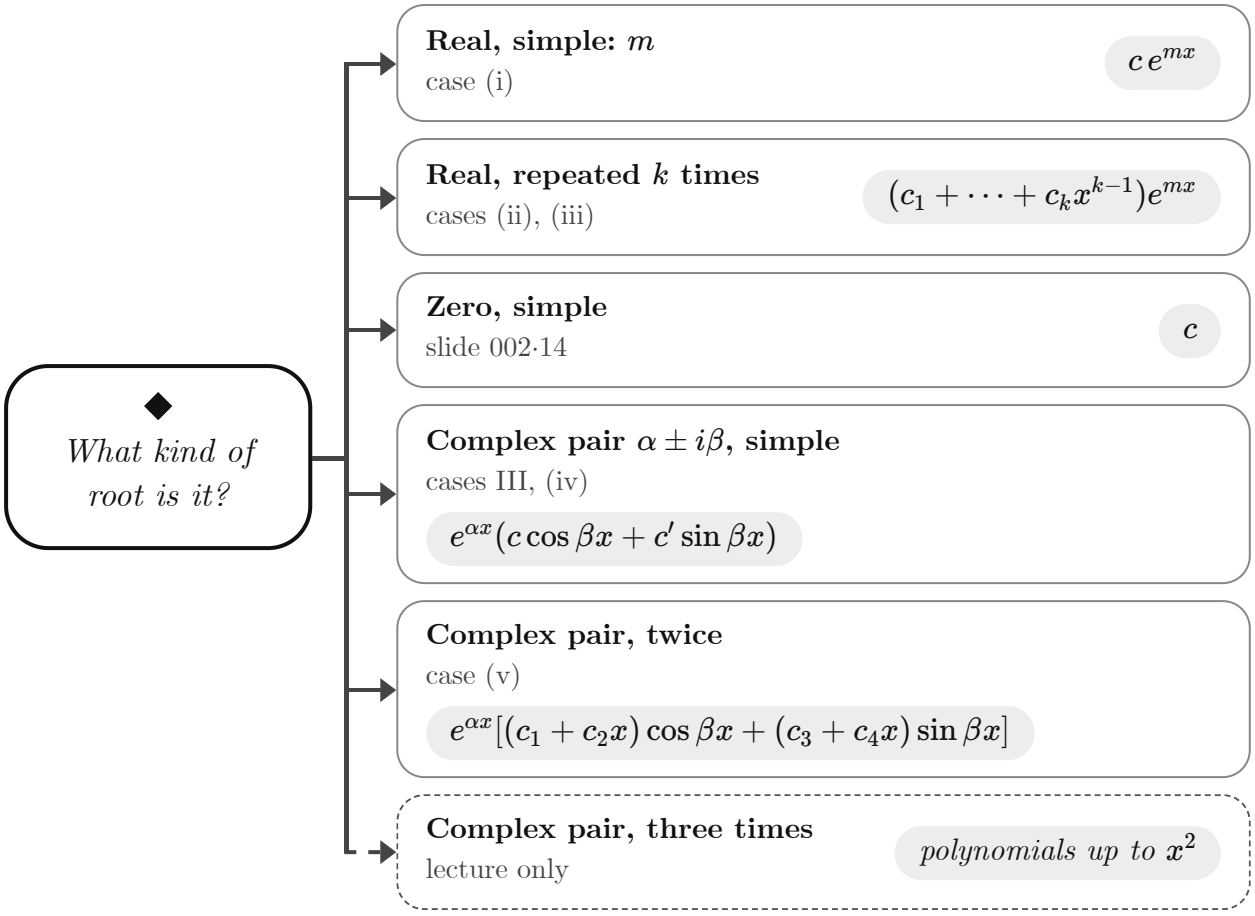


Figure G3.2. A repeated complex pair: $\cos 5x$ against $x\cos 5x$. ESA ★★

ADDED FOR CLARITY CW U2 P7 · S002·13

How to read it. For CW P7 the roots $\pm 5i$ occur twice. A simple pair would give the bounded $\cos 5x$ (blue); the repeat adds $x\cos 5x$ (vermilion dashed), whose amplitude grows linearly between the grey lines $y = \pm x$, although $\alpha = 0$.

Flowcharts



Remark. Add the blocks for all the roots. The dashed route is from the lecture (T002 00:19:14), not the slides.

Figure F3.1. Which CF block does each root give? ESA ★★★

S002·11 · S002·12 · S002·13 · S002·14 · T002 00:19:14

The five order-n cases of slides 002·11–13

Case	Roots	General solution
(i)	all real, distinct	$c_1 e^{m_1 x} + \cdots + c_n e^{m_n x}$
(ii)	all real, equal to m	$(c_1 + c_2 x + \cdots + c_n x^{n-1})e^{mx}$
(iii)	four equal to m, rest distinct	$(c_1 + c_2 x + c_3 x^2 + c_4 x^3)e^{mx} + c_5 e^{m_5 x} + \ldots$
(iv)	$\alpha \pm i\beta$, $\gamma \pm i\delta$, rest distinct	$e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x) + e^{\gamma x}(c_3\cos\delta x + c_4\sin\delta x) + \ldots$
(v)	$\alpha \pm i\beta$ twice, rest distinct	$e^{\alpha x}[(c_1 + c_2 x)\cos\beta x + (c_3 + c_4 x)\sin\beta x] + \ldots$

S002·11 · S002·12 · S002·13

Important points and why

1. A root repeated k times gives $(c_1 + c_2x + \cdots + c_kx^{k-1})e^{mx}$: the highest power is x^{k-1}, not x^k. ESA ★★★ S002·12 · T002 00:15:15
2. Build the CF block by block, one block per distinct root (or conjugate pair), then number the constants $c_1, c_2, \ldots$ straight through. ESA ★★★ T002 00:16:37 · T002 00:18:09
3. A repeated complex pair multiplies both $\cos\beta x$ and $\sin\beta x$ by a polynomial: $(c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x$. ESA ★★ S002·13 · T002 00:19:14
4. Factorise higher-degree AEs by trying $m = \pm1, \pm2, \pm\frac{1}{2}, \ldots$ and synthetic division; biquadratics like $m^4 + 13m^2 + 36$ factor as quadratics in m^2. ESA ★★ ADDED FOR CLARITY CW U2 P6 · CW U2 P9

Exam trap. For a triple root write up to x^2, not x^3. The slide's case (iii) has four equal roots, hence up to x^3. S002·12 · T002 00:15:15

Exam trap. In case (iv) the second complex pair takes new constants c_3, c_4. The lecture notes that an older version of the slide reused c_1, c_2 (T002 00:18:09); the current slide 002·13 is correct. S002·13 · T002 00:18:09

Worked examples

W3.1 · $y''' + 4y' = 0$ (slide 002·14) ESA ★★★

GIVEN Solve $\frac{d^3y}{dx^3} + 4\frac{dy}{dx} = 0$.

STEP 1 $(D^3 + 4D)y = 0$; AE $m^3 + 4m = 0$, i.e. $m(m^2 + 4) = 0$.

STEP 2 $m = 0$ (real) and $m = \pm2i$ ($\alpha = 0$, $\beta = 2$).

STEP 3 $y = c_1e^{0x} + e^{0x}(c_2\cos 2x + c_3\sin 2x)$.

ANSWER $\boxed{y = c_1 + c_2\cos 2x + c_3\sin 2x}$

CHECK sympy agrees; three constants for a third-order equation.

S002·14 · T002 00:20:09 · T002 00:21:14 · T002 00:21:38

W3.2 · Two complex pairs: $\frac{d^4x}{dt^4} + 4x = 0$ (CW P5) ESA ★★

GIVEN Solve $\frac{d^4x}{dt^4} + 4x = 0$.

STEP 1 AE $m^4 + 4 = 0$. Write $m^4 + 4 = (m^2 + 2)^2 - 4m^2 = (m^2 - 2m + 2)(m^2 + 2m + 2)$.

STEP 2 $m = 1 \pm i$ and $m = -1 \pm i$: two pairs of complex roots, case (iv).

ANSWER $\boxed{x = e^{-t}(c_1\cos t + c_2\sin t) + e^{t}(c_3\cos t + c_4\sin t)}$

CHECK CW answer agrees; sympy agrees.

CW U2 P5

W3.3 · A quartic with four real roots (CW P6) ESA ★★

GIVEN Solve $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$.

STEP 1 AE $4m^4 - 8m^3 - 7m^2 + 11m + 6 = 0$. $m = -1$: $4 + 8 - 7 - 11 + 6 = 0$ ✓; $m = 2$: $64 - 64 - 28 + 22 + 6 = 0$ ✓.

STEP 2 Divide out $(m + 1)(m - 2) = m^2 - m - 2$: quotient $4m^2 - 4m - 3 = (2m - 3)(2m + 1)$.

STEP 3 $m = -1,\ 2,\ \frac{3}{2},\ -\frac{1}{2}$: all real and distinct.

ANSWER $\boxed{y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{\frac{3}{2}x} + c_4 e^{-\frac{1}{2}x}}$

CHECK CW answer agrees; sympy roots and dsolve agree.

CW U2 P6

W3.4 · Repeated complex pair: $y^{iv} + 50y'' + 625y = 0$ (CW P7) ESA ★★

GIVEN Solve $y^{iv} + 50y'' + 625y = 0$.

STEP 1 AE $m^4 + 50m^2 + 625 = (m^2 + 25)^2 = 0$.

STEP 2 $m = \pm 5i$, each twice: case (v) with $\alpha = 0$, $\beta = 5$.

ANSWER $\boxed{y = (c_1 + c_2 x)\cos 5x + (c_3 + c_4 x)\sin 5x}$

CHECK CW answer agrees; sympy agrees. Figure G3.2.

CW U2 P7 · S002·13

W3.5 · CF of $(D-1)^2(D+1)^2y = X$ (ESA Jan-24 s4 2c, CW P22) ESA ★★★

GIVEN Find the CF of $[(D - 1)^2(D + 1)^2]y = \sin^2 \frac{x}{2} + 2^{-x} + x$.

STEP 1 AE $(m - 1)^2(m + 1)^2 = 0$: $m = 1$ twice and $m = -1$ twice.

STEP 2 One block of E3.2 for each double root.

ANSWER $\boxed{y_c = (c_1 + c_2 x)e^x + (c_3 + c_4 x)e^{-x}}$

CHECK Agrees with the CW P22 answer; the PI is in the class 005 pack.

ESA Jan-24 s4 2c · CW U2 P22

W3.6 · CF of $(D^3 - 2D^2 + D)y = X$ (ESA Jan-24 s3 2c, CW P18) ESA ★★★

GIVEN Find the CF of $(D^3 - 2D^2 + D)y = x^2e^{-x} + \sin^2 x$.

STEP 1 AE $m^3 - 2m^2 + m = m(m-1)^2 = 0$.

STEP 2 $m = 0$ (simple) gives c_1; $m = 1$ (double) gives $(c_2 + c_3x)e^x$.

ANSWER $\boxed{y_c = c_1 + (c_2 + c_3x)e^x}$

CHECK Agrees with the CW P18 answer; the PI is in the class 006 pack.

ESA Jan-24 s3 2c · CW U2 P18

Past-paper questions

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q2.c (CF part) · solved in W3.5

Solve $[(D-1)^2(D+1)^2]y = \sin^2\frac{x}{2} + 2^{-x} + x$. (CF: two double roots.)

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q2.c (CF part) · solved in W3.6

Solve $(D^3 - 2D^2 + D)y = x^2e^{-x} + \sin^2 x$. (CF: a zero root and a double root.)

SHORT-NOTE RECAP

Order n: n roots, n constants. Each simple real root: ce^{mx}; a k-fold real root: $(c_1 + \cdots + c_kx^{k-1})e^{mx}$; each complex pair: $e^{\alpha x}(c\cos\beta x + c'\sin\beta x)$; a complex pair twice: $e^{\alpha x}[(c_1 + c_2x)\cos\beta x + (c_3 + c_4x)\sin\beta x]$; a zero root: a constant. Add the blocks. $y''' + 4y' = 0$: $c_1 + c_2\cos 2x + c_3\sin 2x$.

4 Solving problems, with initial conditions

Every problem follows the same steps: write the AE, factorise it, classify the roots, write the CF. When initial (or boundary) conditions are given, differentiate the CF, substitute the conditions and solve the linear system for $c_1, c_2, \ldots$. Class-work problems 1–9 cover every case. No ESA paper asks a homogeneous equation on its own, but every non-homogeneous ESA question starts with this CF.

Graphs and diagrams

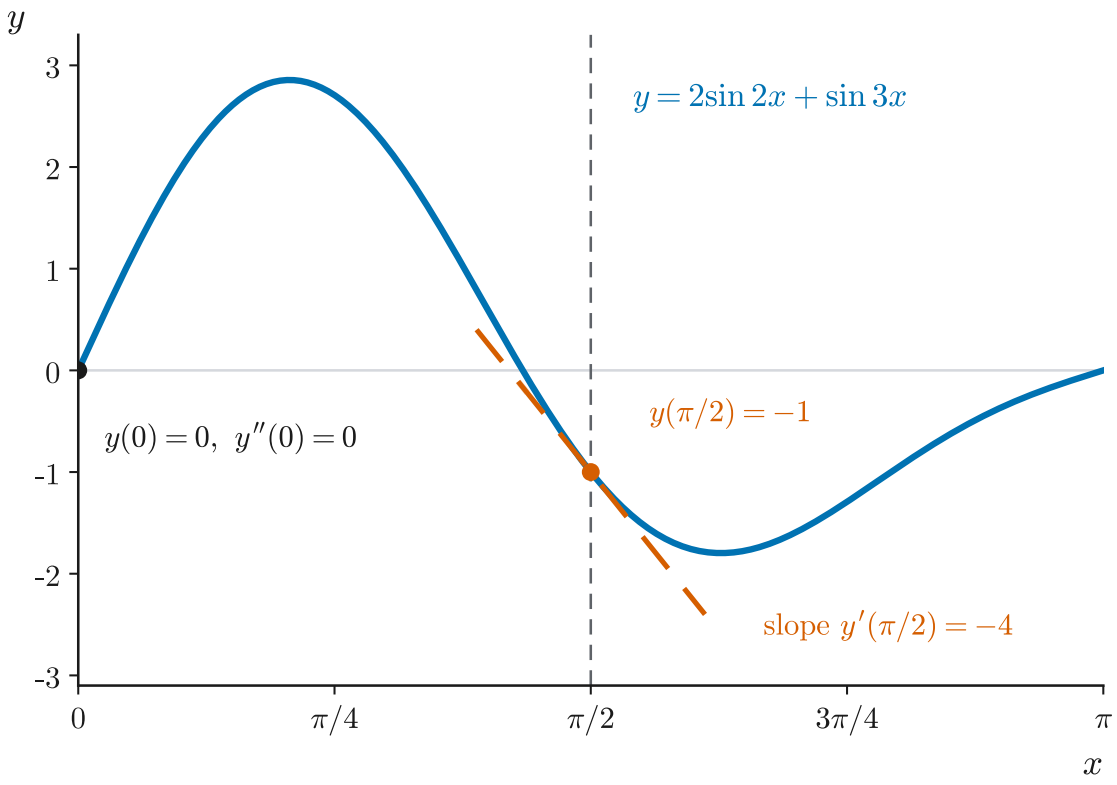
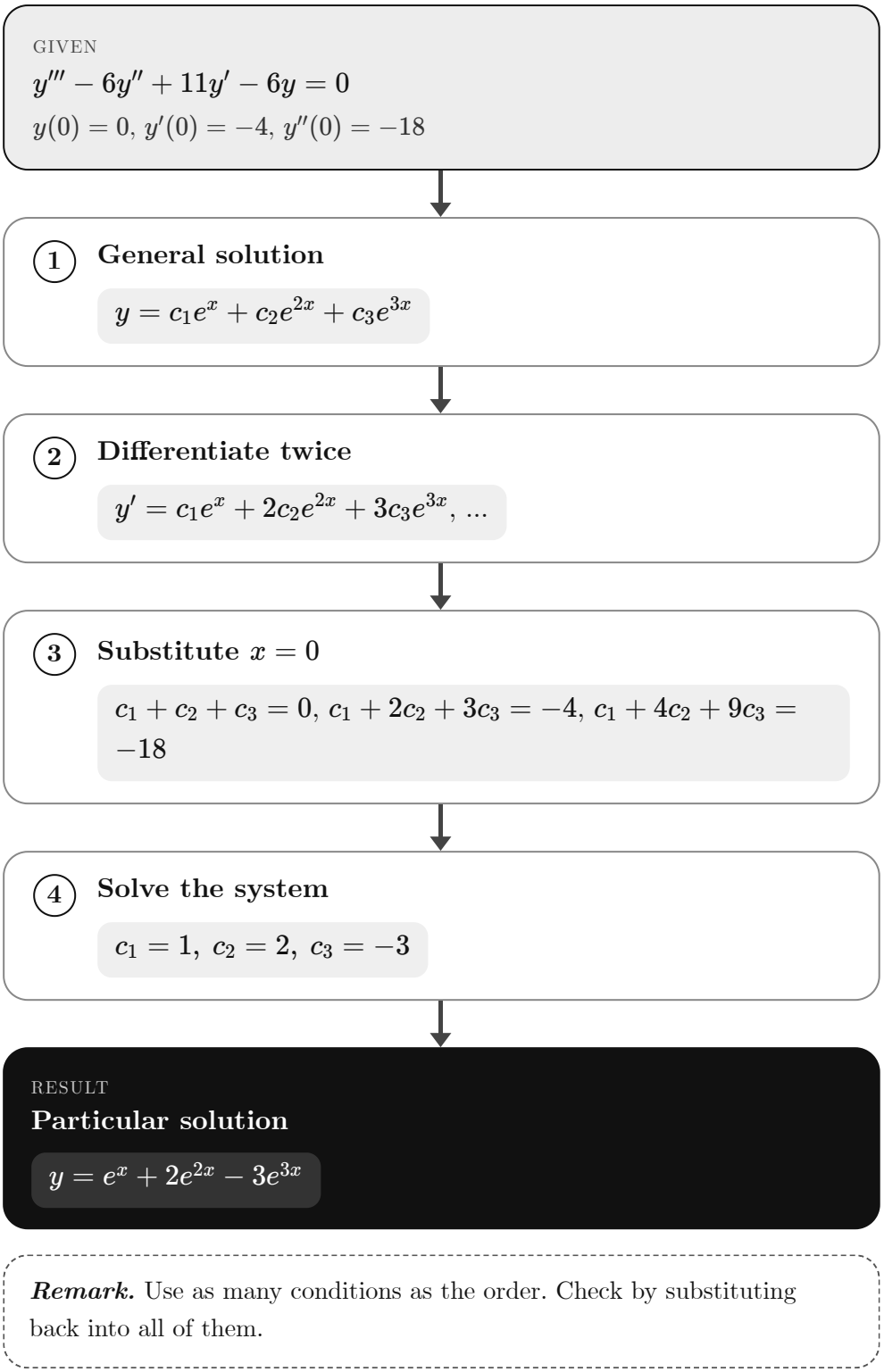


Figure G4.1. The solution of CW P9 meets its four conditions. ESA ★★

ADDED FOR CLARITY CW U2 P9

How to read it. $y = 2\sin 2x + \sin 3x$ (blue) passes through the origin with $y''(0) = 0$ (black dot) and through $(\frac{\pi}{2}, -1)$ (vermilion dot) with slope -4 (vermilion dashed tangent). Four conditions fix the four constants of a fourth-order equation.

Flowcharts



Remark. Use as many conditions as the order. Check by substituting back into all of them.

Figure F4.1. Constants from initial conditions (CW P8) ESA ★★

CW U2 P8

Class-work problems 1–9 at a glance

CW	Roots	Case	Answer (verified)
P1	$6, -2$	distinct	$c_1 e^{6x} + c_2 e^{-2x}$
P2	$-2, -2$	equal	$(c_1 + c_2 x)e^{-2x}$
P3	$2 \pm i$	complex + IVP	$e^{2x}(2\cos x - 5\sin x)$
P4	$\frac{3}{2}, \frac{1}{2}$	distinct + IVP	$\frac{1}{2}(5e^{3x/2} - 3e^{x/2})$
P5	$1 \pm i,\ -1 \pm i$	two complex pairs	$e^{-t}(c_1\cos t + c_2 \sin t) + e^{t}(c_3 \cos t + c_4 \sin t)$
P6	$-1, 2, \frac{3}{2}, -\frac{1}{2}$	distinct	$c_1 e^{-x} + c_2 e^{2x} + c_3 e^{3x/2} + c_4 e^{-x/2}$
P7	$\pm 5i$ twice	repeated complex	$(c_1 + c_2 x)\cos 5x + (c_3 + c_4 x)\sin 5x$
P8	$1, 2, 3$	distinct + IVP	$e^x + 2e^{2x} - 3e^{3x}$
P9	$\pm 2i, \pm 3i$	complex + BVP	$2\sin 2x + \sin 3x$

CW U2 P1 · CW U2 P5 · CW U2 P9

The CF inside every distinct Unit 2 ESA question

Paper, Q	AE (after substitution)	Roots	CF
Dec-23 3a (= May-24 3b, Jun-24 2a)	$(m-2)^2$	$2, 2$	$(c_1 + c_2 x)e^{2x}$
Dec-23 3b	$\theta^2 + 1$, $z = \log(1+x)$	$\pm i$	$c_1 \cos z + c_2 \sin z$
Dec-23 3c	$m^2 + 3m + 2$	$-1, -2$	$c_1 e^{-x} + c_2 e^{-2x}$
Jul-23 3a (= May-24 3a)	$4(\theta-2)^2$, $z = \log(2x+1)$	$2, 2$	$(c_1 + c_2 z)e^{2z}$
Jul-23 3b	$m^2 - 5m + 6$	$2, 3$	$c_1 e^{2x} + c_2 e^{3x}$
Jul-23 3c (= Jun-24 3a)	$(m+3)^2$	$-3, -3$	$(c_1 + c_2 x)e^{-3x}$
May-24 3c	$(m-1)^2$	$1, 1$	$(c_1 + c_2 x)e^{x}$
Jan-24 s1 2d	$m^2 + 4$	$\pm 2i$	$c_1 \cos 2x + c_2 \sin 2x$
Jan-24 s1 3a	$m^2 - 2m + 2$	$1 \pm i$	$e^x(c_1 \cos x + c_2 \sin x)$
Jan-24 s1 3c	$4(\theta-3)(\theta+1)$, $z = \log(2x+1)$	$3, -1$	$c_1 e^{3z} + c_2 e^{-z}$
Jan-24 s2 2d	$(m-1)(m-2)(m-3)$	$1, 2, 3$	$c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$
Jan-24 s2 3a	$m^2 + 2m + 2$	$-1 \pm i$	$e^{-x}(c_1 \cos x + c_2 \sin x)$
Jan-24 s2 3c	$(\theta-1)^2$, $z = \log(x+2)$	$1, 1$	$(c_1 + c_2 z)e^{z}$

Paper, Q	AE (after substitution)	Roots	CF
Jan-24 s3 2c	$m(m-1)^2$	$0, 1, 1$	$c_1 + (c_2 + c_3 x)e^x$
Jan-24 s3 3b (LCR)	$m^2 + 40m + 625$	$-20 \pm 15i$	$e^{-20t}(c_1 \cos 15t + c_2 \sin 15t)$
Jan-24 s3 3d (= s4 3c)	$(\theta - 2)(\theta - 3)$, $z = \log x$	$2, 3$	$c_1 x^2 + c_2 x^3$
Jan-24 s4 2c	$(m-1)^2(m+1)^2$	$1, 1, -1, -1$	$(c_1 + c_2 x)e^x + (c_3 + c_4 x)e^{-x}$
Jan-24 s4 3b (LCR)	$m^2 + 20m + 500$	$-10 \pm 20i$	$e^{-10t}(c_1 \cos 20t + c_2 \sin 20t)$
Jun-24 3b	$2(\theta - 1)(2\theta + 1)$, $z = \log(2x - 1)$	$1, -\frac{1}{2}$	$c_1 e^z + c_2 e^{-z/2}$

ADDED FOR CLARITY

ESA Dec-23 3a · ESA Dec-23 3b · ESA Dec-23 3c · ESA Jul-23 3a · ESA Jul-23 3b · ESA Jul-23 3c · ESA May-24 3c · ESA Jan-24 s1 2d · ESA Jan-24 s1 3a · ESA Jan-24 s1 3c · ESA Jan-24 s2 2d · ESA Jan-24 s2 3a · ESA Jan-24 s2 3c · ESA Jan-24 s3 2c · ESA Jan-24 s3 3b · ESA Jan-24 s3 3d · ESA Jan-24 s4 2c · ESA Jan-24 s4 3b · ESA Jun-24 3b

Important points and why

1. With initial conditions, find the general solution first, then differentiate it as often as the conditions need, then substitute. ESA ★★ CW U2 P3 · CW U2 P8
2. Conditions need not be at $x = 0$: CW P9 gives $y(\frac{\pi}{2})$ and $y'(\frac{\pi}{2})$ (a boundary-value problem). ESA ★★ CW U2 P9
3. The independent variable need not be x: CW P5 is $\frac{d^4x}{dt^4} + 4x = 0$, so the CF is in t. ESA ★ CW U2 P5

Exam trap. Differentiate the whole general solution (product rule on $e^{2x}\cos x$) before substituting $x = 0$; $y'(0) \neq c_2$ in CW P3. ADDED FOR CLARITY CW U2 P3

Worked examples

W4.1 · Complex roots with initial values (CW P3) — ESA ★★

GIVEN Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 5y = 0$, $y(0) = 2$, $y'(0) = -1$.

STEP 1 AE $m^2 - 4m + 5 = 0$, $m = 2 \pm i$: $y = e^{2x}(c_1 \cos x + c_2 \sin x)$.

STEP 2 $y(0) = c_1 = 2$.

STEP 3 $y' = e^{2x}\big[(2c_1 + c_2)\cos x + (2c_2 - c_1)\sin x\big]$, so $y'(0) = 2c_1 + c_2 = -1$ and $c_2 = -5$.

ANSWER $\boxed{y = e^{2x}(2\cos x - 5\sin x)}$

CHECK CW answer agrees; sympy dsolve with ics agrees; $y(0) = 2$, $y'(0) = -1$ ✓.

CW U2 P3

W4.2 · Fractional roots with initial values (CW P4, homework) ESA ★★

GIVEN Solve $4y'' - 8y' + 3y = 0$, $y(0) = 1$, $y'(0) = 3$.

STEP 1 AE $4m^2 - 8m + 3 = (2m-1)(2m-3) = 0$, $m = \frac{1}{2}, \frac{3}{2}$: $y = c_1 e^{\frac{3}{2}x} + c_2 e^{\frac{1}{2}x}$.

STEP 2 $y(0)$: $c_1 + c_2 = 1$. $y'(0)$: $\frac{3}{2}c_1 + \frac{1}{2}c_2 = 3$.

STEP 3 Subtract half the first: $c_1 = \frac{5}{2}$, $c_2 = -\frac{3}{2}$.

ANSWER $\boxed{y = \frac{1}{2}\left(5e^{\frac{3x}{2}} - 3e^{\frac{x}{2}}\right)}$

CHECK CW answer agrees; sympy agrees; $y(0) = 1$, $y'(0) = 3$ ✓.

CW U2 P4

W4.3 · Third order with initial values (CW P8) ESA ★★★

GIVEN Solve $y''' - 6y'' + 11y' - 6y = 0$, $y(0) = 0$, $y'(0) = -4$, $y''(0) = -18$.

STEP 1 AE $m^3 - 6m^2 + 11m - 6 = (m-1)(m-2)(m-3) = 0$: $y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x}$.

STEP 2 $c_1 + c_2 + c_3 = 0$; $c_1 + 2c_2 + 3c_3 = -4$; $c_1 + 4c_2 + 9c_3 = -18$.

STEP 3 Subtracting: $c_2 + 2c_3 = -4$ and $3c_2 + 8c_3 = -18$, so $c_3 = -3$, $c_2 = 2$, $c_1 = 1$.

ANSWER $\boxed{y = e^x + 2e^{2x} - 3e^{3x}}$

CHECK CW answer agrees; sympy agrees; all three conditions ✓. The same operator is the CF of ESA Jan-24 s2 2d.

CW U2 P8 · ESA Jan-24 s2 2d

W4.4 · Boundary values at $\pi/2$ (CW P9, homework) ESA ★★

GIVEN Solve $y'''' + 13y'' + 36y = 0$, $y(0) = 0$, $y''(0) = 0$, $y(\frac{\pi}{2}) = -1$, $y'(\frac{\pi}{2}) = -4$.

STEP 1 AE $m^4 + 13m^2 + 36 = (m^2+4)(m^2+9) = 0$, $m = \pm 2i, \pm 3i$: $y = c_1 \cos 2x + c_2 \sin 2x + c_3 \cos 3x + c_4 \sin 3x$.

STEP 2 $y(0) = c_1 + c_3 = 0$; $y''(0) = -4c_1 - 9c_3 = 0$, so $c_1 = c_3 = 0$.

STEP 3 $y(\frac{\pi}{2}) = c_2 \sin \pi + c_4 \sin \frac{3\pi}{2} = -c_4 = -1$, so $c_4 = 1$.

STEP 4 $y' = 2c_2 \cos 2x + 3c_4 \cos 3x$; $y'(\frac{\pi}{2}) = -2c_2 + 0 = -4$, so $c_2 = 2$.

ANSWER $\boxed{y = 2\sin 2x + \sin 3x}$

CHECK CW answer agrees; sympy dsolve with the four conditions agrees. Figure G4.1.

CW U2 P9

Past-paper questions

ESA Jan 2024 · 5 marks · UE23MA141A set 2, Q2.d (CF part) · solved in W4.3

Solve $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$. (CF: roots 1, 2, 3, as in CW P8.)

SHORT-NOTE RECAP

Initial values: write the general solution, differentiate, substitute, solve for the constants. CW: P3 $e^{2x}(2\cos x - 5\sin x)$; P4 $\frac{1}{2}(5e^{3x/2} - 3e^{x/2})$; P8 $e^{x} + 2e^{2x} - 3e^{3x}$; P9 $2\sin 2x + \sin 3x$. All nine CW answers check out.

A Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $D^2 = \frac{d^2}{dx^2}$	E1.2
$f(D)$	Linear differential operator	$D^n + k_1 D^{n-1} + \cdots + k_n$	E1.2
$f(m)$	Auxiliary polynomial	$f(D)$ with D replaced by m	E1.2
m, $m_1, \ldots, m_n$	Roots of the AE	Exponents of the trial solution e^{mx}	E2.1–E3.5
$k_1, \ldots, k_n$	Constant coefficients	Given numbers in the DE	E1.2, E2.4
$c_1, \ldots, c_n$	Arbitrary constants	n of them for order n; fixed by initial conditions	all
α, β	Real and imaginary parts	Of a complex root $\alpha \pm i\beta$ ($\beta > 0$)	E2.3, E3.4, E3.5
γ, δ	Second complex pair	$\gamma \pm i\delta$ in case (iv)	E3.4
Δ	Discriminant	$k_1^2 - 4k_2$ (added for clarity)	F2.1
k	Multiplicity	Times a root is repeated. Not the coefficients k_1, k_2	E3.2
θ, z	Legendre/Cauchy substitution	$z = \log(ax+b)$, $\theta = \frac{d}{dz}$ (classes 011; table C4.2 only)	C4.2

B Sources, coverage and gaps

Source	Path	Use
Deck 002 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/002_Solutions_of_Homogeneous_DE_by_introducing_Complementary_Functions__c193c841b234d834.*	15 slides; content on 4–14 (every equation read from the rendered images)
Transcript 002 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.*	22 min; slides 3–14
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Unit 2 problems 1–9 (with answers; page 5 rendered to settle superscripts); CF parts of problems 18 and 22
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	No homogeneous question; the CF of every Unit 2 question (table C4.2), via the shared ESA map
Video 002 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/002_Rules_for_finding_the_Complementary_Function_Problems__4a39dee530243163.mp4	Not opened: every slide is readable and the transcript has no low-confidence segment

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No paper asks a homogeneous equation on its own (0 of 8), so no question is about this class alone. But every Unit 2 ESA question is non-homogeneous and needs a CF first: 24 occurrences in the 8 papers, all of them (table C4.2). The CF cases appear as follows over the distinct questions: real distinct roots 6, real repeated roots 7 (including two double pairs in Jan 2024 set 4 Q2.c and a zero root with a double root in set 3 Q2.c), complex roots 6 (of 19 distinct questions). Stars follow that count: the three basic cases ★★★; two complex pairs, a repeated complex pair and initial conditions ★★ (class-work only); a complex pair repeated three times ★ (lecture only).

Known gaps, conflicts and corrections

1. No slide or class-work answer in this class is wrong: all four slide examples and all nine CW answers (P1–P9, including the homework problems P4 and P9 and the initial and boundary values of P3, P4, P8, P9) were recomputed with sympy dsolve and checkodesol.
2. The lecture says "alpha is 2 and beta is 2" for $2 \pm 3i$ before correcting to 3 (T002 00:12:13); slide 002·9 is right.
3. The lecture says $\frac{d^3y}{dx^3}$ is replaced by "m square" (T002 00:07:49); it is m^3. Recorded as an exam trap.
4. The lecture says an earlier version of slide 002·13 used c_1, c_2 for the second complex pair of case (iv) (T002 00:18:09). The current slide uses c_3, c_4, which is correct.

5. The complex pair repeated three times (E3.6) is in the lecture only (T002 00:19:14–00:19:45), not on the slides.
6. Slide 002·8's heading drops the word "homogeneous"; slide 002·12's case (iii) has an extra outer bracket around the whole solution. Neither changes the maths.
7. The lecture's "complex roots always occur in pairs" holds because the coefficients are real; added in the points.
8. The deck has no worked problems beyond four illustrations; the problems come from the class-work sheet. The deck has no graphs.
9. ESA: no homogeneous equation is asked on its own in the 8 papers. The shared ESA map gives 24 Unit 2 occurrences; its own table lists 19 distinct questions, although its summary line says 17.
10. Bucket session numbers are not contiguous (decks say SESSION 1–11; bucket 009 = session 7, 011 VoP = 8, 011 Legendre = 9, 014 = 10, 015 = 11; 007, 008, 010, 012, 013 do not exist).
11. Transcript recognition errors were corrected silently: "rules"/"loads"/"load" → roots, "d2 by b dx2" → $\frac{d^2y}{dx^2}$, "f of b" → $f(D)$, "a to the power mx" → e^{mx}, "dy by dx by u" → by m, "LB"/"LT"/"LD" → LDE, "n thirder" → n-th order, "Dq y by dx q"/"mq" → $\frac{d^3y}{dx^3}/m^3$, "arbitrar" → arbitrary, "diffusion differential" → differential, "r minus"/"pi beta" → $\pm/i\beta$, "sine beta s" → $\sin\beta x$, "c phi"/"m phi"/"mc" → $c_5/m_5/m_6$, "complex load" → complex root, "plus or minus divide" → $\pm 2i$, "sine to x" → $\sin 2x$.

Coverage

Source	Citations	Slides cited (slide × citations)
S002	68	4×1, 5×4, 6×4, 7×7, 8×9, 9×11, 10×2, 11×4, 12×10, 13×10, 14×6
CW	25	
ESA	27	
T002	63	

Slides 002·1, 002·2 and 002·15 are title, session and closing slides; slide 002·3 lists the class content. The lecture (22 min) covers every content slide 4–14 in order. The deck has no class-work problems; they come from the class-work sheet (problems 1–9, all solved here). Bucket session numbers are not contiguous: the decks say SESSION 1–11, but the bucket numbers them 001–006, 009, 011 (two decks), 014 and 015; 007, 008, 010, 012 and 013 do not exist. This deck is SESSION 2 in both numberings.